

ASSIGNMENT NO. 10

PROBLEM STATEMENT:

Write a program in C to simulate a simple calculator that performs basic arithmetic operations such as addition, subtraction, multiplication, and division. The calculator should also perform special operations like computing x^y (power) and $x!$ (factorial).

OBJECTIVE:

- To understand the use of switch-case statements in C.
- To perform basic arithmetic and special mathematical operations.
- To enhance problem-solving skills using menu-driven programs.

THEORY:

A calculator is a basic computing device used to perform arithmetic operations. In C programming, a simple calculator can be implemented using conditional or switch-case statements. Arithmetic operations like addition, subtraction, multiplication, and division are fundamental operations. Special operations such as power (x^y) and factorial ($x!$) require loops for repeated calculations.

PLATFORM:

Linux

CODE:

```
#include <stdio.h>
#include <math.h>

unsigned long long factorial(int n) {
    if (n < 0) return 0;
    if (n == 0 || n == 1) return 1;
    unsigned long long fact = 1;
    for (int i = 2; i <= n; i++) {
        fact *= i;
    }
    return fact;
}

int main() {
    double num1, num2, result;
    int choice;

    printf("=== Simple Calculator ===\n");
    printf("1. Addition (+)\n");
    printf("2. Subtraction (-)\n");
    printf("3. Multiplication (*)\n");
    printf("4. Division (/)\n");
    printf("5. Power (x^y)\n");
    printf("6. Factorial (x!)\n");
    printf("7. Exit\n");
    printf("Enter your choice (1-7): ");

    scanf("%d", &choice);

    switch (choice) {
        case 1:
            printf("Enter two numbers: ");
            scanf("%lf %lf", &num1, &num2);
            result = num1 + num2;
            printf("%.2lf + %.2lf = %.2lf\n", num1, num2, result);
            break;
```

```

case 2:
    printf("Enter two numbers: ");
    scanf("%lf %lf", &num1, &num2);
    result = num1 - num2;
    printf("%.2lf - %.2lf = %.2lf\n", num1, num2, result);
    break;

case 3:
    printf("Enter two numbers: ");
    scanf("%lf %lf", &num1, &num2);
    result = num1 * num2;
    printf("%.2lf * %.2lf = %.2lf\n", num1, num2, result);
    break;

case 4:
    printf("Enter two numbers: ");
    scanf("%lf %lf", &num1, &num2);
    if (num2 == 0) {
        printf("Error: Division by zero!\n");
    } else {
        result = num1 / num2;
        printf("%.2lf / %.2lf = %.2lf\n", num1, num2, result);
    }
    break;

case 5:
    printf("Enter base and exponent: ");
    scanf("%lf %lf", &num1, &num2);
    result = pow(num1, num2);
    printf("%.2lf ^ %.2lf = %.2lf\n", num1, num2, result);
    break;

```

```

case 6:
    printf("Enter a non-negative integer: ");
    scanf("%lf", &num1);
    if (num1 != (int)num1 || num1 < 0) {
        printf("Error: Factorial only for non-negative integers!\n");
    } else {
        result = factorial((int)num1);
        printf("%d! = %.0lf\n", (int)num1, result);
    }
    break;

case 7:
    printf("Goodbye!\n");
    return 0;

default:
    printf("Invalid choice! Please try again.\n");
}

return 0;
}

```

OUTPUT:

```
=== Simple Calculator ===
1. Addition (+)
2. Subtraction (-)
3. Multiplication (*)
4. Division (/)
5. Power (x^y)
6. Factorial (x!)
7. Exit
Enter your choice (1-7): 1
Enter two numbers: 22 33
22.00 + 33.00 = 55.00
surabhi_datta@DESKTOP-5SCQMHG:~$ ./output_10
=== Simple Calculator ===
1. Addition (+)
2. Subtraction (-)
3. Multiplication (*)
4. Division (/)
5. Power (x^y)
6. Factorial (x!)
7. Exit
Enter your choice (1-7): 2
Enter two numbers: 7 4
7.00 - 4.00 = 3.00
surabhi_datta@DESKTOP-5SCQMHG:~$ ./output_10
=== Simple Calculator ===
1. Addition (+)
2. Subtraction (-)
3. Multiplication (*)
4. Division (/)
5. Power (x^y)
6. Factorial (x!)
7. Exit
Enter your choice (1-7): 3
Enter two numbers: 44 10
44.00 * 10.00 = 440.00
```

```

=== Simple Calculator ===
1. Addition (+)
2. Subtraction (-)
3. Multiplication (*)
4. Division (/)
5. Power (x^y)
6. Factorial (x!)
7. Exit
Enter your choice (1-7): 4
Enter two numbers: 99 99
99.00 / 99.00 = 1.00
surabhi_datta@DESKTOP-5SCQMHG:~$ ./output_10
=== Simple Calculator ===
1. Addition (+)
2. Subtraction (-)
3. Multiplication (*)
4. Division (/)
5. Power (x^y)
6. Factorial (x!)
7. Exit
Enter your choice (1-7): 5
Enter base and exponent: 2 8
2.00 ^ 8.00 = 256.00
surabhi_datta@DESKTOP-5SCQMHG:~$ ./output_10
=== Simple Calculator ===
1. Addition (+)
2. Subtraction (-)
3. Multiplication (*)
4. Division (/)
5. Power (x^y)
6. Factorial (x!)
7. Exit
Enter your choice (1-7): 6
Enter a non-negative integer: 5
5! = 120

```

```

=== Simple Calculator ===
1. Addition (+)
2. Subtraction (-)
3. Multiplication (*)
4. Division (/)
5. Power (x^y)
6. Factorial (x!)
7. Exit
Enter your choice (1-7): 7
Goodbye!

```

CONCLUSION:

Thus, a simple calculator program was successfully implemented using C programming. The program demonstrates the use of switch-case statements, loops, and arithmetic operations.

